

Q1.

Choose words from this list to complete the sentences below.

balanced

electricity

gravity

joules

magnetism

newtons

When you drop something it falls.

This is because it is pulled to the Earth by _____

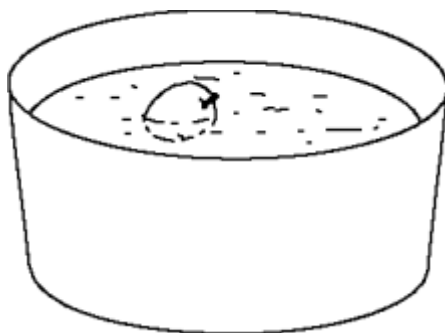
We measure forces in units called _____

When a falling object reaches the ground, it stops moving.

This means that the forces acting on it are now _____

(Total 3 marks)

Q2.



In a science lesson, some children float an apple on some water.

One of the children says:

“The apple is not moving. That means that there cannot be any forces acting on it.”

Do you agree?

Explain your answer as fully as you can.

(Total 3 marks)

Q3.

- (a) **Figure 1** shows an aircraft flying at a constant velocity and at a constant height above the ground.

Figure 1



Complete the free body diagram in **Figure 2** to show the other two forces acting on the aircraft.

Figure 2



(2)

- (b) A small aircraft accelerated down a runway at 4.0 m/s^2

The aircraft started from rest and reached a speed of 34 m/s just before take-off.

Calculate the distance the aircraft travelled while accelerating.

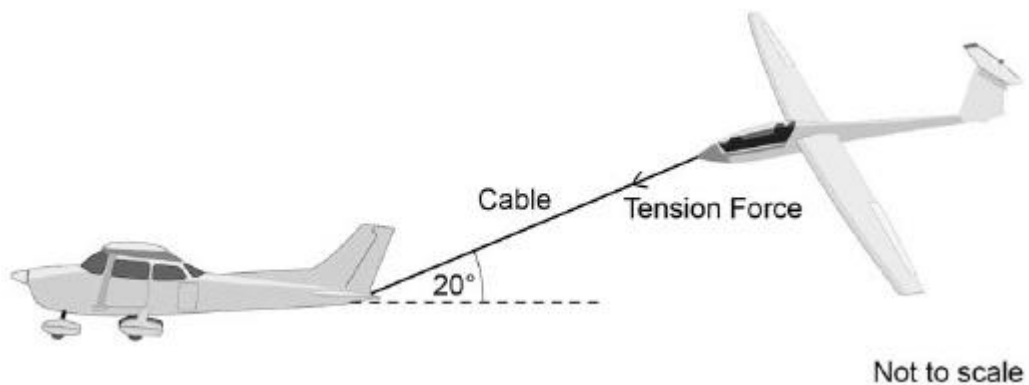
Give your answer to 2 significant figures.

Distance = _____ m

(4)

- (c) **Figure 3** shows the small aircraft being used to tow a glider.

Figure 3



The tension force in the cable can be resolved into a horizontal component and a vertical component.

The tension in the cable is 2000 N

The cable makes an angle of 20° with the horizontal.

Draw a vector diagram to determine the magnitude of the two components of the tension force in the cable.

Magnitude of the horizontal component = _____ N

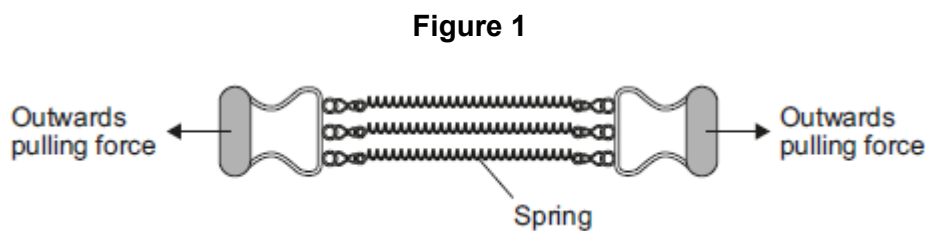
Magnitude of the vertical component = _____ N

(1)

(Total 10 marks)

Q4.

Figure 1 shows an exercise device called a chest expander. The three springs are identical.



A person pulls outwards on the handles and does work to stretch the springs.

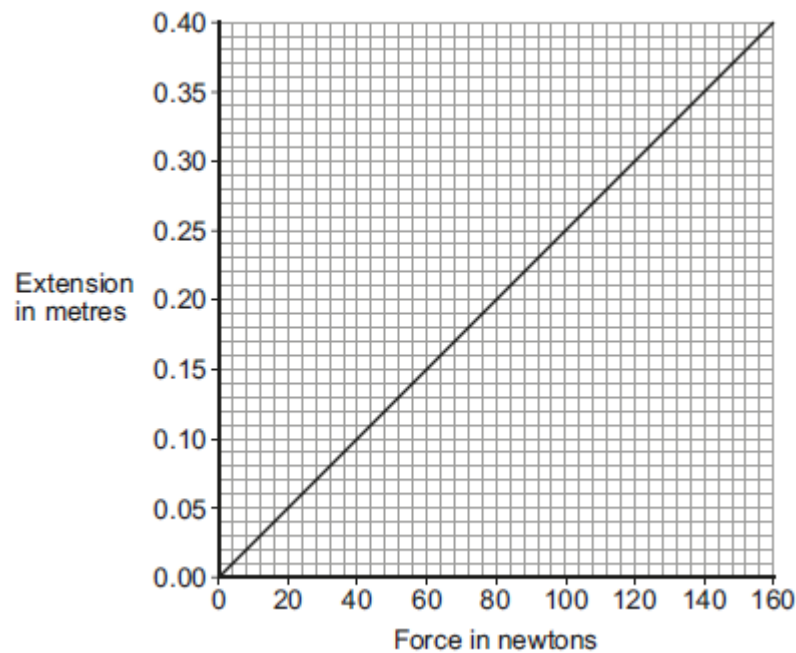
(a) Complete the following sentence.

When the springs are stretched _____ energy is stored in the springs.

(1)

(b) **Figure 2** shows how the extension of a single spring from the chest expander depends on the force acting on the spring.

Figure 2



- (i) How can you tell, from **Figure 2**, that the limit of proportionality of the spring has not been exceeded?

(1)

- (ii) Use data from **Figure 2** to calculate the spring constant of the spring. Give the unit.

Spring constant = _____ Unit _____

(3)

- (iii) Three identical resistors joined in parallel in an electrical circuit share the total current in the circuit.

In a similar way, the three springs in the chest expander share the total force exerted.

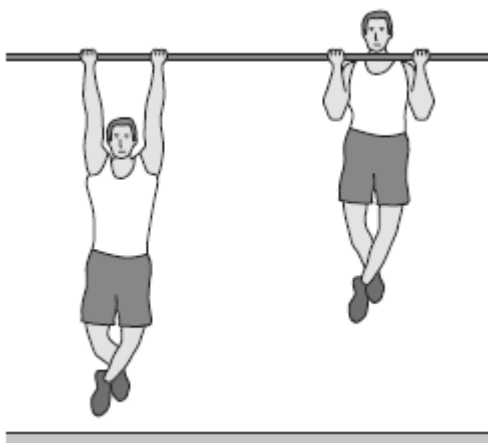
By considering this similarity, use **Figure 2** to determine the total force exerted on the chest expander when each spring is stretched by 0.25 m.

Total force = _____ N

(2)

- (c) The student in **Figure 3** is doing an exercise called a chin-up.

Figure 3



Each time the student does one chin-up he lifts his body 0.40 m vertically upwards.
The mass of the student is 65 kg.
The student is able to do 12 chin-ups in 60 seconds.

Calculate the power developed by the student.

Gravitational field strength = 10 N/kg

Power = _____ W

(3)

(Total 10 marks)

Q5.

A student carried out an investigation to determine the spring constant of a spring.

The table below gives the data obtained by the student.

Force in N	Extension in cm
0	0.0
2	3.5
4	8.0
6	12.5
8	16.0
10	20.0

- (a) Describe a method the student could have used to obtain the data given in the table above.

Your answer should include any cause of inaccuracy in the data.

Your answer may include a labelled diagram.

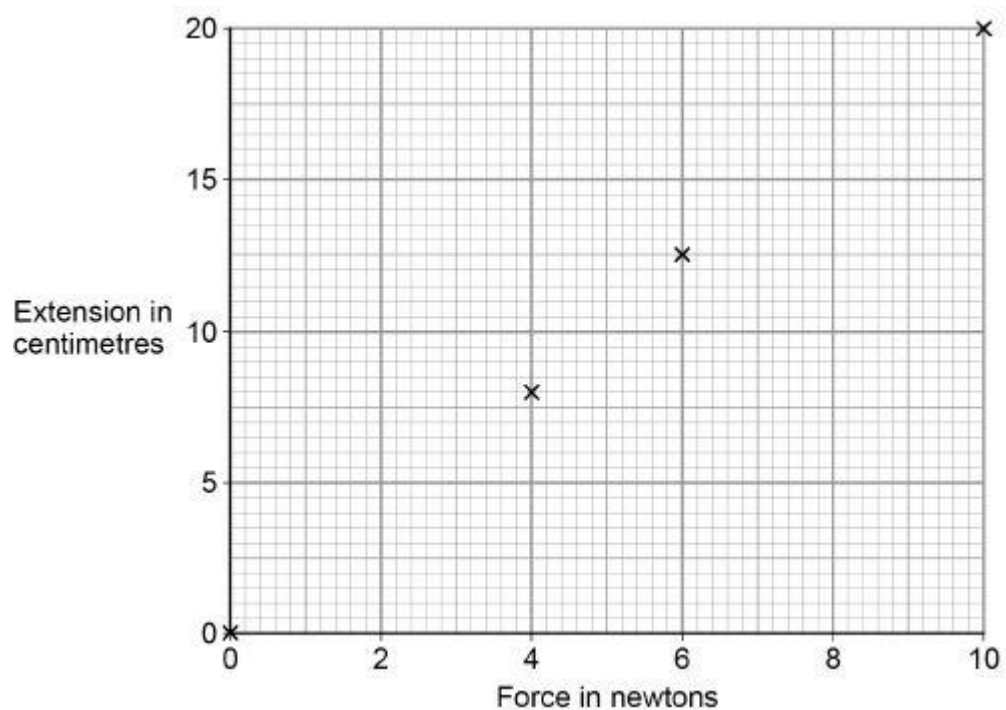
(6)

- (b) The student measured the extension for five different forces rather than just measuring the extension for one force.

Suggest why.

(1)

The diagram below shows some of the data obtained by the student.



- (c) Complete the diagram above by plotting the missing data from the table above.

Draw the line of best fit.

The table above is repeated here to help you answer this question.

Force in N	Extension in cm
0	0.0
2	3.5
4	8.0
6	12.5
8	16.0
10	20.0

(2)

- (d) Write down the equation that links extension, force and spring constant.

(1)

- (e) Calculate the spring constant of the spring that the student used.

Give your answer in newtons per metre.

Spring constant = _____ N/m

(4)

- (f) Hooke's Law states that:
'The extension of an elastic object is directly proportional to the force applied, provided the limit of proportionality is not exceeded.'

The student concluded that over the range of force used, the spring obeyed Hooke's Law.

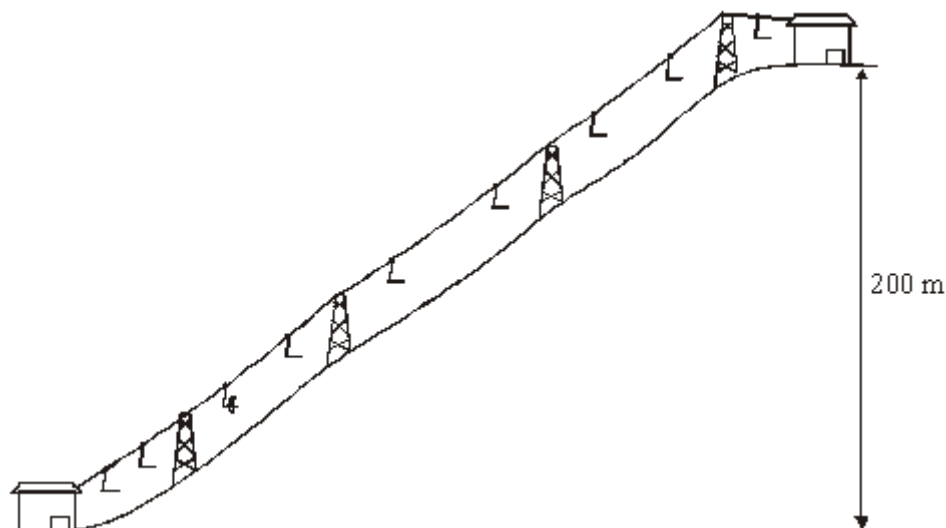
Explain how the data supports the student's conclusion.

(2)

(Total 16 marks)

Q6.

- (a) A chair lift carries two skiers, Greg and Jill, to the top of a ski slope. Greg weighs 700 N and Jill weighs 500 N.



- (i) Write down the equation that links distance moved, force applied and work done.

(1)

- (ii) Calculate the work done to lift Greg and Jill through a vertical height of 200 m. Show clearly how you work out your answer and give the unit.

work done = _____

(3)

- (b) The chair takes 5 minutes to move from the bottom to the top of the ski slope.

Calculate the power required to lift Greg and Jill to the top of the ski slope. Show clearly how you work out your answer.

power = _____ watts

(2)

- (c) The chair lift is driven by an electric motor.

- (i) Why would the power output of the electric motor need to be larger than your answer to part (b)?

_____ (1)

- (ii) Complete the following sentence.

When the ski lift is working _____ energy supplied to the motor
is usefully transferred as gravitational _____ energy.

(1)

(Total 8 marks)

Q7.

A car has an oil leak. Every 5 seconds an oil drop falls from the bottom of the car onto the road.

- (a) What force causes the oil drop to fall towards the road?

_____ (1)

- (b) The diagram shows the spacing of the oil drops left on the road during part of a journey



Describe the motion of the car as it moves from **A** to **B**.

Explain the reason for your answer.

(3)

- (c) When the brakes are applied, a braking force slows down and stops the car.

- (i) The size of the braking force affects the braking distance of the car.

State **one** other factor that affects the braking distance of the car.

_____ (1)

- (ii) A braking force of 3 kN is used to slow down and stop the car in a distance of 25 m.

Calculate the work done by the brakes to stop the car and give the unit.

Work done = _____

(3)

(Total 8 marks)

Mark schemes

Q1.

gravity
newtons
balanced

each for 1 mark

[3]

Q2.

ideas that
gravity/weight (downwards)
upwards/opposite force of water **or** upthrust
forces are balanced

any three for 1 mark each

(N.B. All these ideas may be included in a short response)

(If no marks gained but candidate makes reference to forces, award 1 mark)

[3]

Q3.

- (a) arrow vertically down – same size as lift – labelled weight
judge by eye

1

arrow to the left – same size as drag - labelled thrust
judge by eye

two correct arrows without labels gains 1 mark

1

- (b) $34^2 - (0^2) = 2 \times 4.0 \times s$

1

$$\frac{34 \times 34}{8} = s$$

1

$$s = 144.5$$

1

$$s = 140 \text{ (2 sig figs)}$$

an answer of 140 scores 4 marks

an answer of 144.5 scores 3 marks

1

- (c) tension force drawn to a suitable scale and in correct direction

1

triangle completed showing correct components

1

scale used to determine both component forces

1

horizontal component = 1900 N

vertical component = 680 N

allow 1850 to 1925 inclusive

allow 660 to 700 inclusive

1

[10]

Q4.

(a) elastic potential

1

(b) (i) line is straight

accept line does not curve

1

(ii) 400

*allow 1 mark for correct substitution of any pair of numbers
correctly taken from the graph e.g. $160 = k \times 0.40$*

2

newtons per metre **or** N/m

if symbols are used they must be correct

1

(iii) 300

allow 1 mark for correctly obtaining force on 1 spring = 100N

2

(c) 52

allow 2 marks for calculating change in gpe for 1 chin-up as 260 (J) or for 12 chin-ups as 3120 (J)

an answer 4.3 gains 2 marks

allow 1 mark for correct substitution into gpe equation ie $gpe = 65 \times 10 \times 0.4 (\times 12)$

or

correct use of power equation with an incorrect value for energy transferred

3

[10]

Q5.

(a) **Level 3:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

5–6

Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.

3–4

Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.

No relevant content

0

Indicative content

set up a clamp stand with a clamp

hang the spring from the clamp

use a second clamp and boss to fix a (half) metre ruler alongside the spring

record the metre ruler reading that is level with the bottom of the spring

hang a 2 N weight from the bottom of the spring

record the new position of the bottom of the spring

calculate the extension of the spring

measure the extension of the spring

add further weights to the spring so the force increases 2 N at a time up to 10 N

for each new force record the position of the bottom of the spring and calculate / measure the extension

possible source of inaccuracy

not fixing the ruler in position but simply holding the ruler next to the spring

not clamping the ruler vertical

misjudging the position of the bottom of the spring

parallax error

allow any other sensible suggestion that could reasonably lead to inaccuracy in the data

allow a description that would increase accuracy

repeating the measurements is insufficient

- (b) to identify any anomalous results
 allow calculate an average for the spring constant

or

to reduce the effect of random error
 allow (more) accurate
 to obtain an average is insufficient
 to be able to draw a graph is insufficient

1

- (c) both points plotted correctly

1

correct line of best fit drawn

to pass through (0,0) and (10,20)

1

- (d) force = spring constant $\times$ extension

allow $F = ke$

1

- (e) extension = 0.2

allow 0.035 / 0.08 / 0.125 / 0.16

1

$$10 = k \times 0.2$$

force value must match extension

this mark may be awarded if e is in cm

1

$$k = \frac{10}{0.2}$$

allow correct transformation of their chosen values

this mark may be awarded if e is in cm

1

$$k = 50$$

an answer 0.5 scores 3 marks

1

an answer of 50 scores 4 marks

- (f) the line is straight

allow the line does not curve

1

and passes through the origin

this mark is dependent on scoring the first mark

allow a correct description of direct

proportionality for 2 marks

ignore the line shows they are directly proportional

1

[16]

Q6.

- (a) (i) work (done) = force (applied) $\times$ distance (moved)

accept $W = F \times s$ or $W = F \times d$

accept  provided subsequent method is correct

1

- (ii) 240 000

*allow 1 mark for correct substitution **or** correct use of 1200*

	(N)		2
	joules		
	accept J		
	do not accept j / Nm		1
(b)	800 (watts)		
	accept 0.8 kW		
	accept their (a)(ii) ÷ 300 correctly evaluated for 2 marks		
	allow 1 mark for correct substitution		
	(a)(ii) ÷ 5 correctly evaluated for 1 mark		2
(c)	(i) any one from:		
	• needs to raise the chair / lift		
	• lifting more than one chair		
	allow lifting more than 2 people		
	implication of a heavier weight		
	• energy transfer to the surroundings		
	correctly qualified		
	accept loss for transfer		
	do not accept motor inefficient		
	do not accept motor gets hot		
	do not accept friction unless the location is specified as external to the motor		1
	(ii) electrical		
	accept electric		
	potential		
	both answers required for the mark		1

[8]

Q7.

(a)	gravitational / gravity / weight		
	do not accept gravitational potential		1
(b)	accelerating		
	accept speed / velocity increases		1
	the distance between the drops increases		1
	but the time between the drops is the same		
	accept the time between drops is (always) 5 seconds		
	accept the drops fall at the same rate		

(c) (i) any **one** from:

- speed / velocity
- (condition of) brakes / road surface / tyres
- weather (conditions)

accept specific examples, eg wet / icy roads

accept mass / weight of car friction is insufficient

reference to any factor affecting thinking distance negates this answer

1

(ii) 75 000

allow 1 mark for correct substitution, ie 3000×25 provided no subsequent step shown

or allow 1 mark for an answer 75

or allow 2 marks for

75 k(+ incorrect unit), eg 75 kN

2

joules / J

do **not** accept j

an answer 75 kJ gains 3 marks

for full marks the unit and numerical answer must be consistent

1

[8]

Examiner reports

Q1.

This recognition item was generally answered well, though not infrequently, joules were thought to be the units of force.

Q2.

Paper 1 Option P

Some candidates had the idea of balanced forces but were unable to clearly describe gravity or upthrust.

Paper 3 Option Q

Able candidates could describe equal forces acting on the apple.

Q4.

- (a) Over half of the students got this correct although some students gave two types of energy rather than one type comprising two words.
- (b)
 - (i) Only just over a third of the students scored this mark. The most common incorrect answers were references to the graph being directly proportional rather than describing the graph line itself as straight or having a constant gradient. Many students said 'the line is constant'.
 - (ii) A quarter of the students scored all three marks with a further half of all students gaining 2 marks. Students could identify correct points from the graph but weaker students struggled with rearranging the formula to change the subject to k . Some students attempted to take values from the graph but didn't read the graph correctly and wrote values down such as 0.5 instead of 0.05. A significant number of students incorrectly used the equation $\text{Work} = F \times d$. Incorrect units were mostly Nm or reversed m/N, or not capitalised n/m, with a variety of other incorrect options eg N, W, J, or nothing. A few students correctly gave the answer as 4 N/cm.
 - (iii) A third of the students scored both marks and a further third scored one mark. Some students correctly extrapolated 100N from the graph but then did not take into account the fact that there were 3 springs so did not multiply by 3 and therefore only scored 1 mark.
- (c) Two thirds of the students scored some marks on this question, although fewer than a fifth scored full marks. Most students were able to calculate the energy required for one chin up, but did not take into account that they were being asked about 12 chin-ups. Some students were able to gain a compensation mark by clearly showing that they were dividing an incorrect energy by time to work out the power.

Q5.

Foundation

- (a) This question related to one of the required practical activities. Students were asked to describe a method and to include causes of inaccuracy. Many students wrote at

length about safety issues, reducing risks, repeating results and removing anomalies which were not asked for in this question. Most students drew a labelled diagram which was helpful, and then described what they had drawn, which wasn't necessary. Only a minority of the students appreciated that the force increased in steps of 2N, up to a maximum of 10N, and an extension was measured for each force. Even fewer mentioned a cause of inaccuracy. Some students appreciated that a range of weights were needed to obtain a table of results and made a relevant comment about measuring the extension. Those students working at level 1 hung a single mass on a spring and measured its length giving no indication how the extension would be determined.

- (b) Students were asked why more than one pair of data was collected. Few students stated that it was so anomalous results could be found or to reduce random error. Most students who gained the mark stated that the accuracy of the results would be increased. Several students suggested that several data points would enable a mean to be calculated but did not state that it would be the mean of the spring constant.
- (c) 43% of students plotted the two missing points correctly and drew a line of best fit that passed between (0, 0) and (10, 20). A significant number only plotted the two points or drew a line without having plotted the missing data. It was evident that many students did not use a ruler to draw a straight line.
- (d) 24% were able to write the correct equation.
- (e) Nearly 70% of the students scored zero. Only 4 % of the students were able to score 4 marks by using the correct equation, choosing relevant data from the table and converting centimetres to metres. The students who correctly found the average force and extension and substituted into the correct equation scored full marks.
- (f) Students were asked to show how a given set of data or a graph demonstrated that two variables were directly proportional. It was expected that students would state that the graph was straight and passed through (0, 0). Only a very small number did so. More of the students gained both marks for stating that as one variable doubled, so did the other. Students were given some credit for stating that every time the force increased by 2N the extension increased by approximately 4cm. This was accepted as an alternative to the graph being linear. Many students just stated that as the force increased so did the extension.

Higher

- (a) Almost all students made a good attempt at describing this investigation but many omitted one of the necessary actions such as adding 2N weights up to 10N or a method of ensuring accuracy. Most of the students drew a diagram of the apparatus which, although often in sketch form, was generally enough to show what was intended. However having done this there was no need to write a lengthy description of the diagram. Most students were aware of the need to determine the extension of the spring, and many described in detail how to do this. Many students quoted a correct range of weights. Students were more likely to describe how to ensure accuracy, rather than pointing out inaccuracies. Most frequent answers included; clamping the ruler vertically, using a pointer on the end of the spring and marking the original length of the spring on the ruler with a piece of tape. All of these were worthy of credit.
- (b) Fewer than 40% of students scored the mark. The most common correct answer was in terms of improving accuracy, the idea of identifying anomalous results was

rare.

- (c) Nearly all students plotted the two data points accurately. However some students did not draw the line with adequate precision and so scored only one mark.
- (d) Just over 75% of students correctly recalled the equation.
- (e) For those students who correctly recalled the equation in part (d), most scored at least 3 marks here. About 35% converted centimetres to metres to score all 4 marks.
- (f) Those students who used the graph to answer this question usually wrote that it was a straight line, but generally failed to write that the line went through the origin. Some students explained that the line would curve if the spring was overstretched. Those students that attempted to answer this in terms of direct proportionality often simply stated that as the force increased, the extension increased. This answer scored zero. Only a small minority correctly explained what is meant by two quantities being directly proportional.

Q6.

Foundation Tier

- (a) (i) A significant number of candidates wrote an appropriate equation, but many errors occurred in the substitution of numbers, some involving an incorrect transformation of their original equation. Although a unit was asked for, many candidates failed to provide one.
- (b) The substitution was completed in most cases, but often without the conversion of minutes into seconds.
- (c) Many candidates realised that the electric motor needed more power in order to lift chairs, heavier people etc. Unfortunately, there were few correct completions of the sentences requiring knowledge of energy transfers. This was despite the term 'electric motor' being used earlier and the inclusion of the word 'gravitational'.

Higher Tier

This question was generally answered well.

- (a) The equation was well known and the calculation generally correct, although a significant number of candidates lost marks for omitting the unit or giving an incorrect one.
- (b) The most common error was to complete the calculation with time left in minutes.
- (c) Many candidates realised that the electric motor needed more power in order to lift chairs, heavier people etc. Unfortunately, there were few correct completions of the sentences requiring knowledge of energy transfers. This was despite the term 'electric motor' being used earlier and the inclusion of the word 'gravitational'.

Q7.

- (a) Most students gave a correct answer. The only common incorrect responses seen were gravitational potential and gravitational potential energy.
- (b) This was well answered with most students gaining at least two marks. Most

students identified that the car was accelerating and that the distance between the oil drops was increasing. Fewer students appreciated that the oil drops falling at regular time intervals was also relevant to confirming that the speed was increasing.

- (c) (i) Over half of the students gave a relevant factor. Students failing to gain the mark usually gave thinking time / distance or a factor that affects thinking distance.
- (ii) Very few students were able to complete all the steps required in order to gain all three marks. Many students failed to change the 3 kN value to 3000 N, but some of these students went on to give an answer of 75 with the unit kJ and so did score full marks. However it was more common for the answer 75 to have no unit attached or an incorrect one, N/m being common.